Research 55: Spatial and Temporal Databases

Scalable Model-Based Management of Massive High Frequency Wind Turbine Data with ModelarDB

September 4, 2025

Abduvoris Abduvakhobov,
Søren Kejser Jensen,
Torben Bach Pedersen,
Christian Thomsen

Aalborg University
Denmark



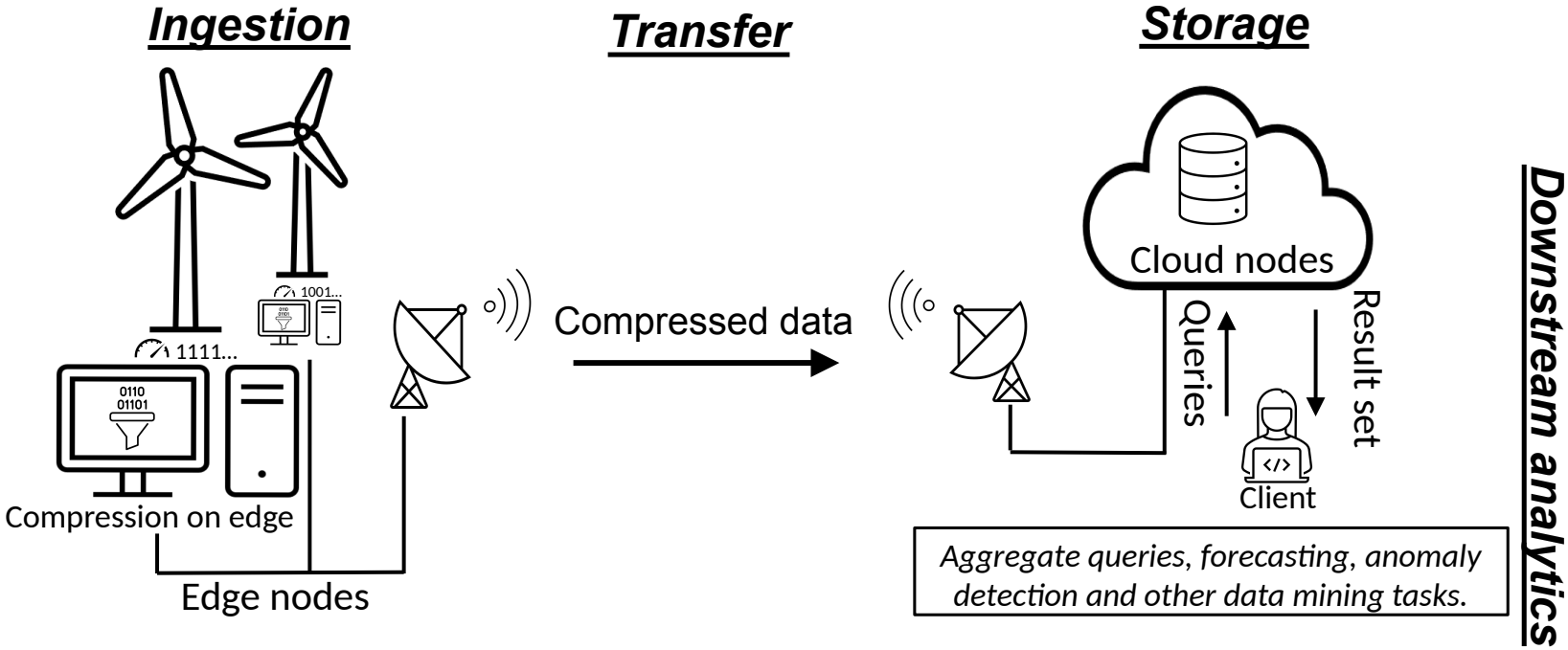
AALBORG UNIVERSITY
DENMARK

Motivation



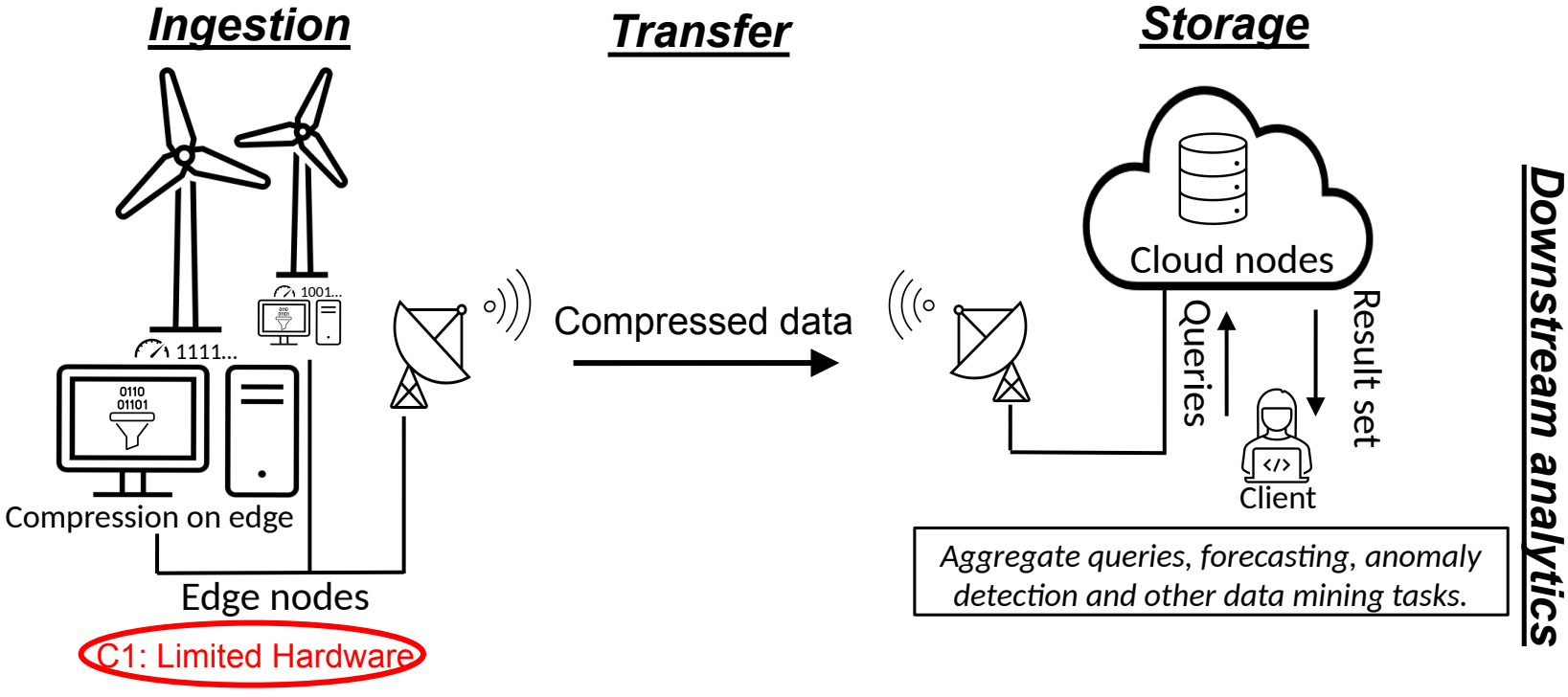
- Wind energy is crucial for green transition efforts.
- Modern wind turbines are monitored by huge amounts of sensors.
- They generate massive amounts of high frequency time series.
- High frequency time series are ingested on the edge and then transferred to the cloud for storage and analytics.
- Due to a number of challenges, this task is currently not possible with all the data.

Challenges with RES Data



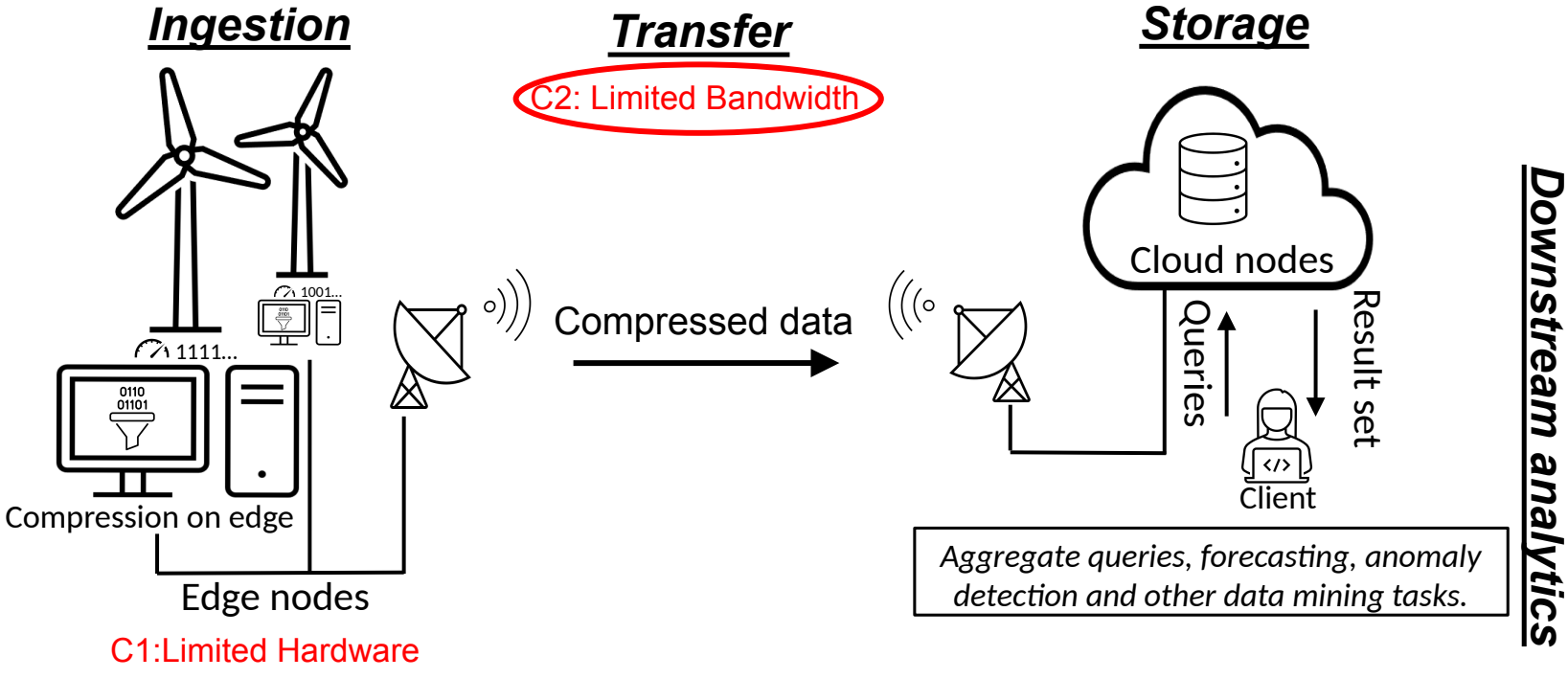
Architecture of edge-to-cloud RES data management.

Challenges with RES Data



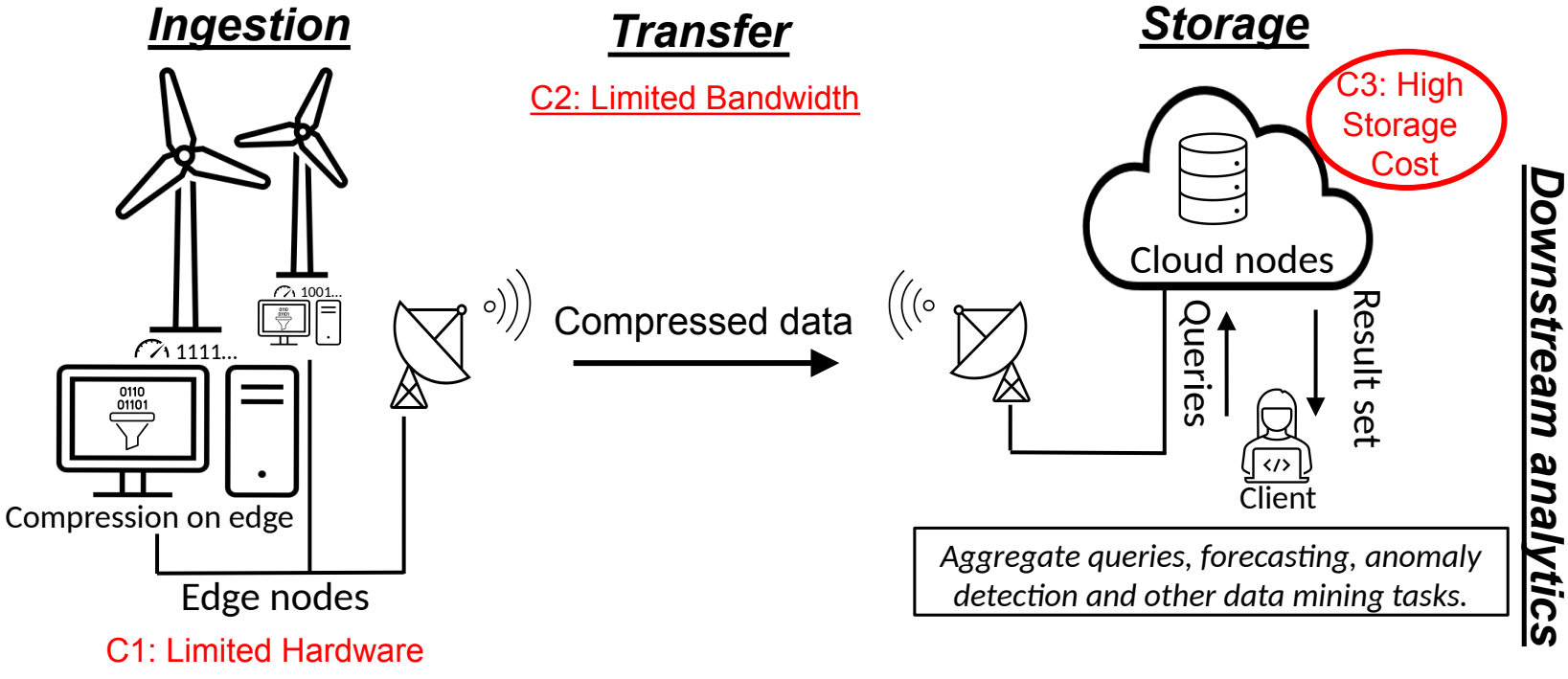
Architecture of edge-to-cloud RES data management.

Challenges with RES Data



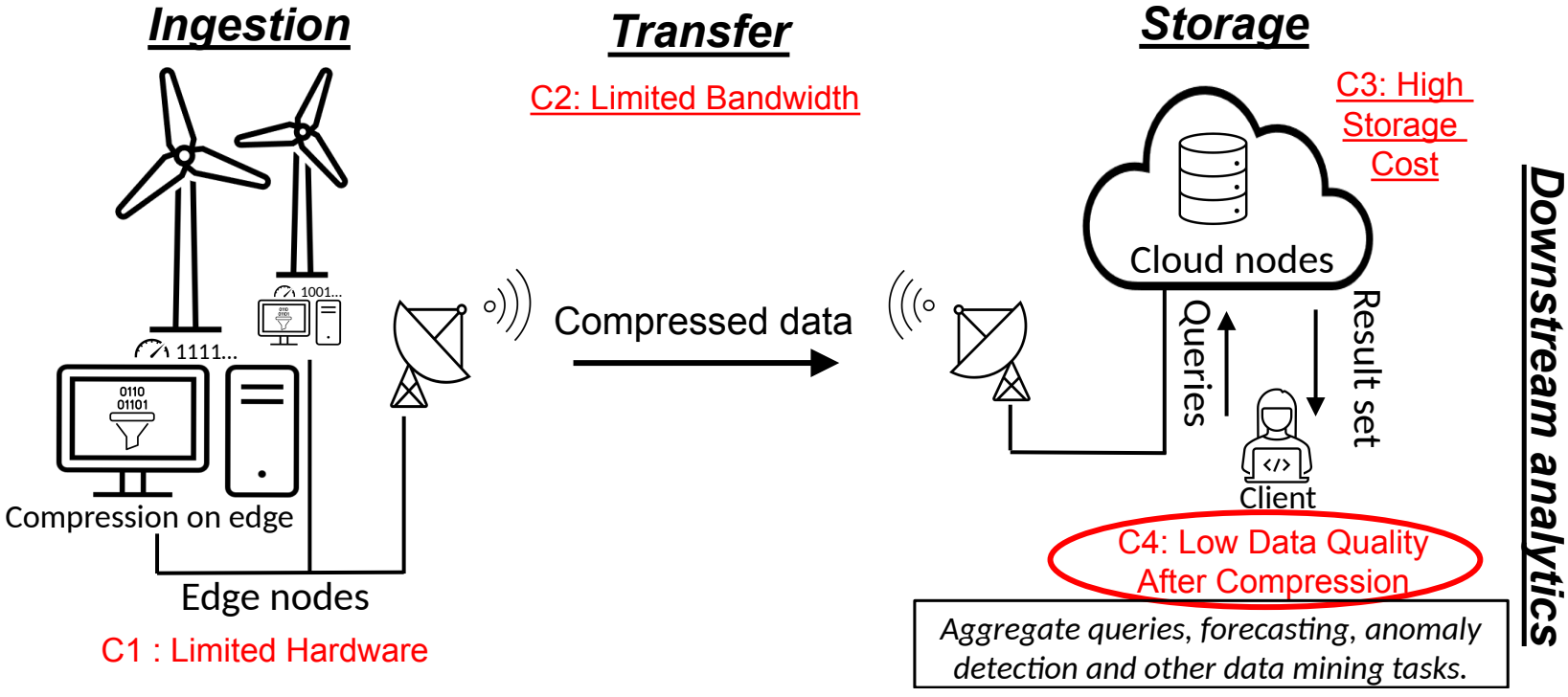
Architecture of edge-to-cloud RES data management.

Challenges with RES Data



Architecture of edge-to-cloud RES data management.

Challenges with RES Data



Architecture of edge-to-cloud RES data management.

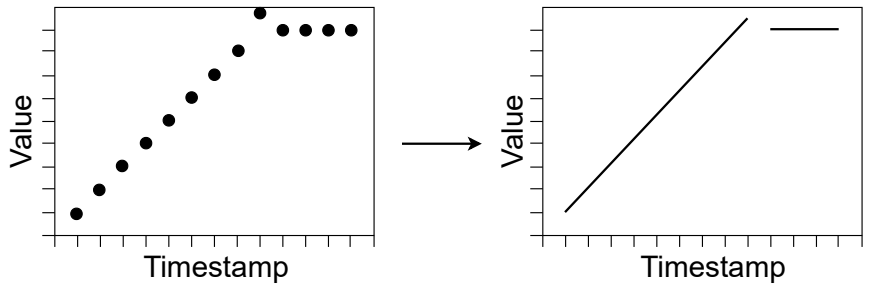
Evaluated Solutions



1. **Lossless Compression (LLC).** Lossless compression using big data formats Apache Parquet and Apache ORC with Snappy compression.
2. **Aggregation (AGG).** Like LLC, but values are aggregated over a fixed window using aggregation function Mean.
3. **IoTDB.** Lossless and lossy compression using TSMS Apache IoTDB.
4. **ModelarDB (MDB).** Lossless and pointwise error-bounded lossy compression using TSMS ModelarDB.

Evaluated Solutions

4. **ModelarDB (MDB).** Stores values as models using *constant*, *linear* and *XOR-based (Gorilla)* functions.
   MDB's Error Bound (ϵ): $error_i \leq \epsilon$ for $error_i = \left| \frac{value_i - value_i'}{value_i} \right|$



MDB's idea of model-based compression.

In this paper, we want to compare MDB to the other solutions.

Evaluation Setup



Real-Life Datasets

- **PCD**. ∼480M data points, SI=150ms, 10 signals.
- **MTD**. ∼258M data points, SI=2s, 6 signals.
- **WTM**. ∼432000 data points, SI=2s, 10 signals, [open access]

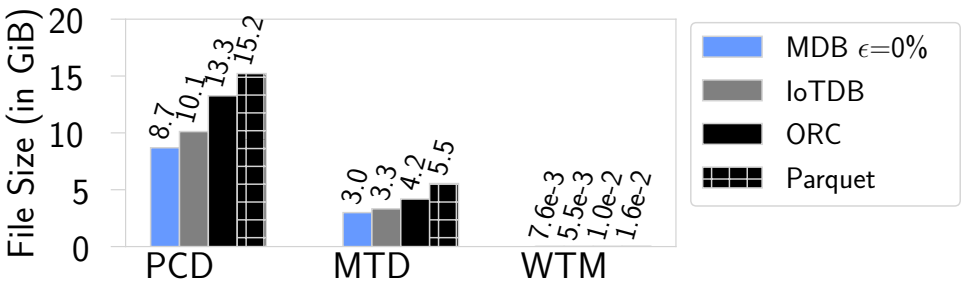
Metrics

- **Compression Factor (CF)**. The ratio of original to compressed.
- **Transfer Efficiency**. The number of values transferred per second.
- **Mean Absolute Percentage Error (MAPE)**.

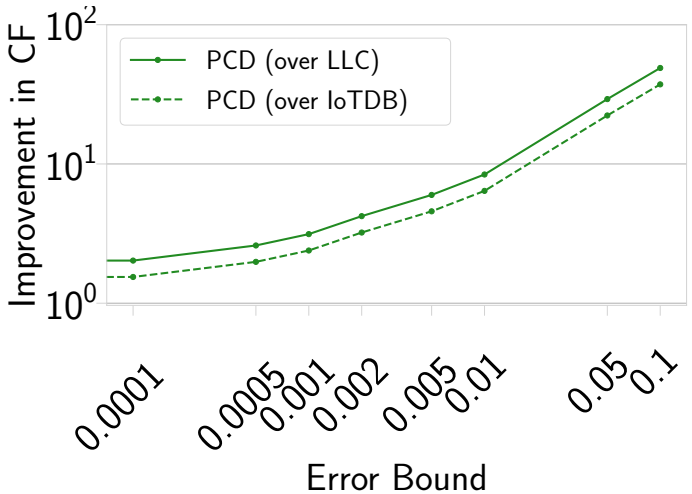
Hardware

- **Edge Node.** Commodity PC running on Linux.
- **Cloud Node.** Server-class PC running on Linux.
- **Network Bandwidth.** 512 kbit/s and 2.5 Mbit/s.

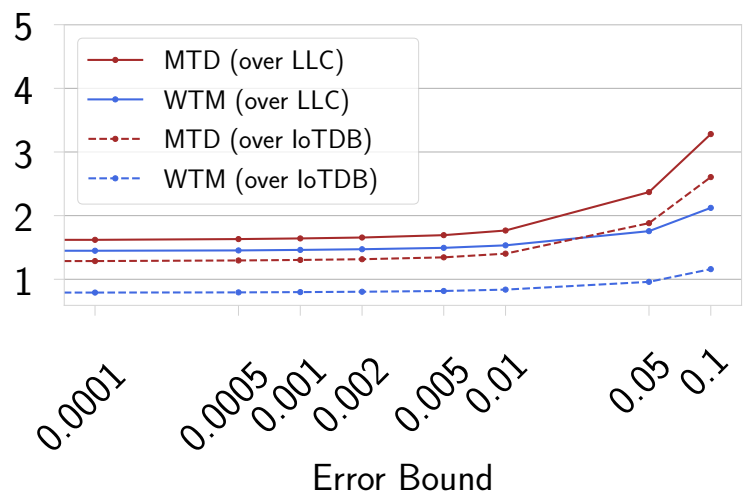
Challenge 1: Limited Hardware



Storage use of ϵ=0% solutions.



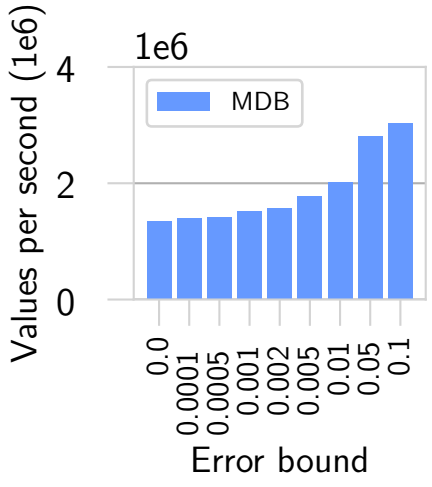
MDB with PCD



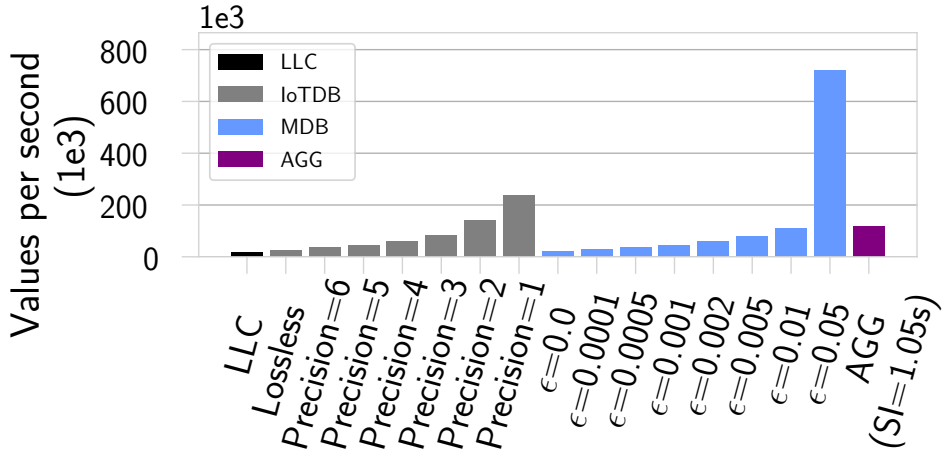
MDB with MTD and WTM

- MDB provides up to *1.5x and 8.4x better compression* than the lossless solutions for ϵ=0% and 1%.

Challenge 2: Limited Bandwidth



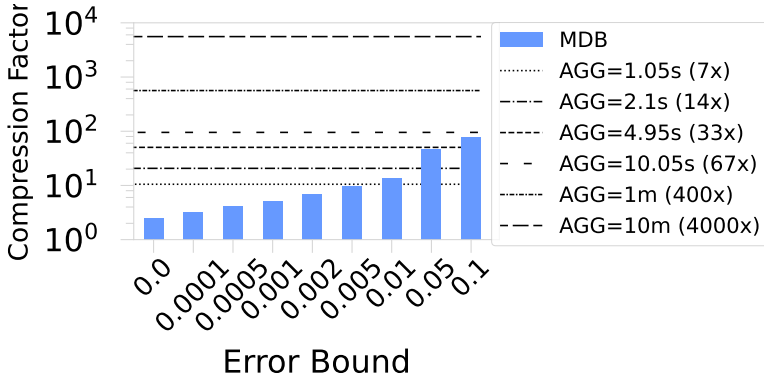
MDB's ingestion on the edge



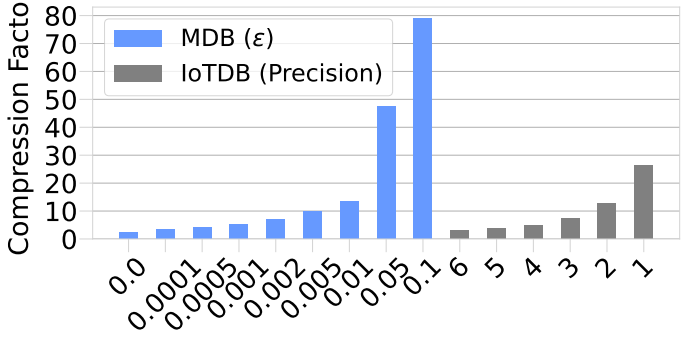
Transfer with 512 kbit/s

- LLC can transfer around *19,000 values per second with a 512 Kbit/s* connection.
- MDB at ϵ=*0%* (i.e., with lossless compression) can transfer *1.2x more values*.
- MDB with ϵ=*1% matches* the transfer efficiency of *AGG by 7 values (7x AGG)* while also having *up to ∼12 orders of magnitude better* data quality.

Challenge 3: High Storage Cost



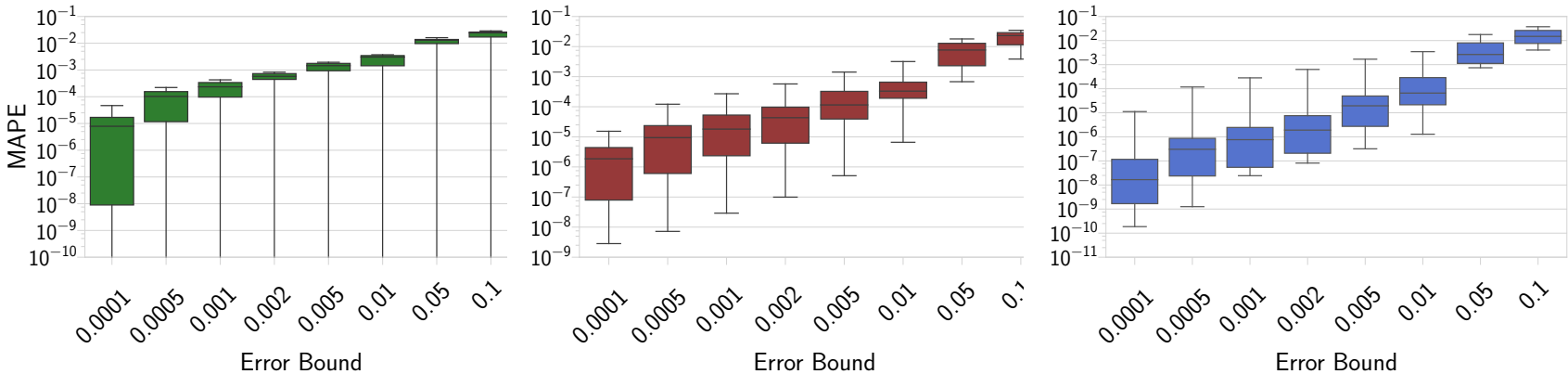
(a) MDB and AGG's compression factor with PCD



(b) MDB and IoTDB's compression factor with PCD

- MDB with ϵ=*0.5% matches 1.05s (7x) AGG.* ϵ=*10% matches 10.05s (67x) AGG.*
- MDB at ϵ=*5% provides ∼1.8x better compression* than IoTDB with *precision=1.*

Challenge 4: Low Data Quality After Compression



(a) PCD (b) MTD (c) WTM

MAPE of PCD, MTD and WTM when compressed with MDB.

- For all datasets, MAPE does not exceed *half of the respective* ϵ.
- With ϵ=0.01%, MDB represents *72-95% of the values losslessly* and with ϵ=10%, it represents *17-34% of the values losslessly*.
- With MDB, aggregate queries generate *much lower relative query error* than the ϵ.

Achieved Insights



Challenge 1 – Limited Hardware:

- MDB's model-based compression *increases compression factor*.
- MDB's model-based compression is the *most effective with very high frequency data*.

Challenge 2 – Limited Bandwidth:

- *Network bandwidth is the main bottleneck* in edge to cloud scenario.
- *Compression on edge* increases transfer throughput to the cloud.
- MDB is the *most effective with the very high frequency dataset* (i.e., PCD).

Achieved Insights



Challenge 3 – High Storage Cost:

- Compared to LLC, MDB's lossless compression *has higher compression factor*.
- Compared to IoTDB, MDB provides *better compression factor and data quality*.
- Compared to AGG, MDB provides *competetive compression factor but much better data quality*.

Challenge 4 – Low Data Quality After Compression:

- MDB provides at least 50% better data quality than the respective ϵ.
- MDB has much lower relative query error than the ϵ.
- AGG and IoTDB (in some instances) generate orders of magnitude higher error than MDB.

Conclusion

	Challenge 1 Limited Hardware	Challenge 2 Limited Bandwidth	Challenge 3 High Storage Cost	Challenge 4 Low Data Quality After Compression
LLC	✓	✗	✗	✓
AGG	✓	✓	✓	✗
IoTDB	✓	✓	(✓)	✓
MDB	✓	✓	✓	✓

Challenges solved by each of the evaluated solutions.

MDB's model-based lossless and error-bounded lossy compression provides the best tradeoff compared to LLC, AGG and IoTDB due to *effective compression on edge, high throughput from the edge to the cloud* and *high data quality after compression*.

Future Work



- Design new model types and improve the model fitting strategy by exploiting ϵ and properties of time series.
- Improve the physical representation and layout of segments on disk for better storage use and query performance.

Useful Links

Paper: Link

Presentation: Link

All ModelarDB Papers: Link

Acknowledgments:

- EU Horizon 2020 projects DEDS (grant: 955895) and MORE (grant: 957345).
- Special thanks to Siemens Gamesa Renewable Energy (SGRE), Emil Hedevang and our anonymous industry partner.

1